

2001

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 13, 17 and 21

Total Marks – 100

Section I Pages 2–24

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 25–31

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

1 Ethene may be converted into poly(ethene).

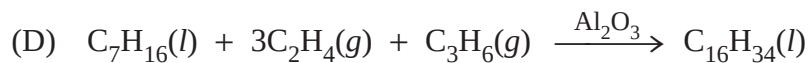
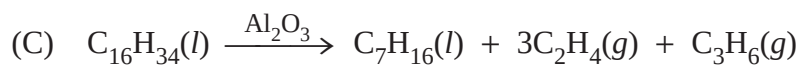
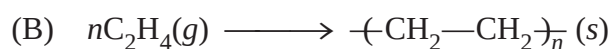
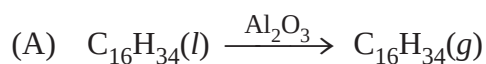
What type of reaction is this?

- (A) Condensation
- (B) Hydrolysis
- (C) Oxidation/reduction
- (D) Polymerisation

2 Which of the following is a major component of biomass?

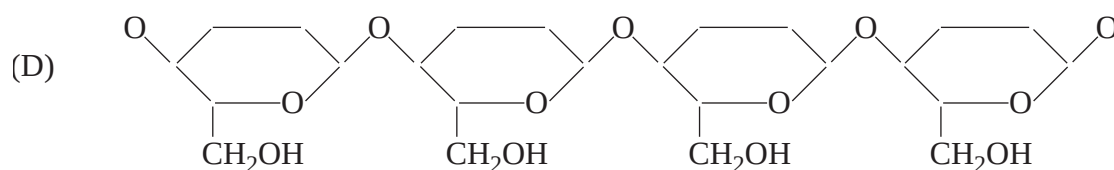
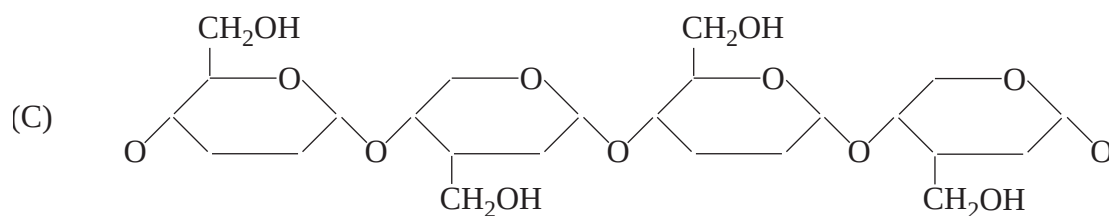
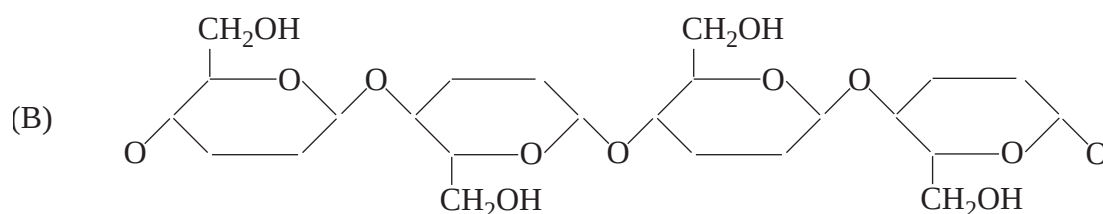
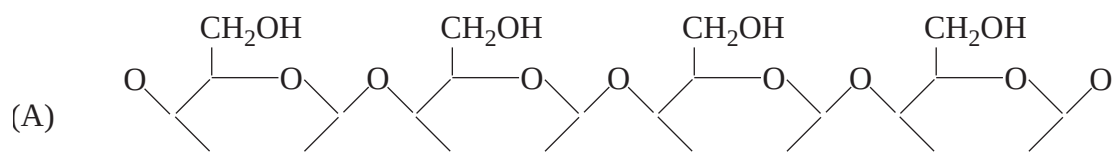
- (A) Cellulose
- (B) Ethanol
- (C) Natural gas
- (D) Oil

3 Which equation best represents catalytic cracking of a petroleum fraction?



- 4 Cellulose is a linear polymer which is a basic structural component of plant cell walls.

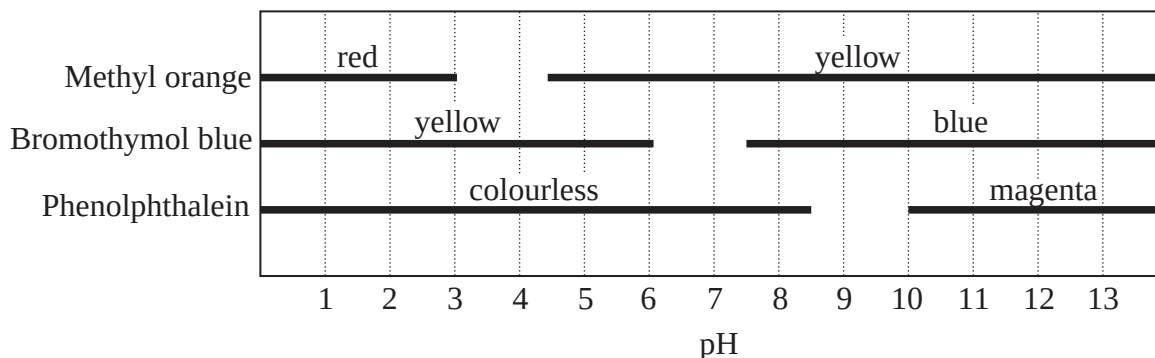
Which is the correct representation of part of a cellulose polymer?



- 5 The pH of unpolluted rainwater is about 6.0. Which substance contributes most to this?

- (A) CO_2
- (B) N_2
- (C) NO_2
- (D) O_3

- 6 The graph shows the colour ranges of the acid–base indicators methyl orange, bromothymol blue and phenolphthalein.



A solution is yellow in methyl orange, blue in bromothymol blue and colourless in phenolphthalein.

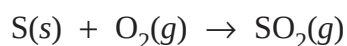
What is the pH range of the solution?

- (A) 4.5 to 6.0
 - (B) 6.0 to 7.5
 - (C) 7.5 to 8.5
 - (D) 8.5 to 10.0
- 7 A group of students produced a red solution by boiling red cabbage leaves in water. When dilute sodium hydroxide was added to the solution, it turned purple. When dilute hydrochloric acid was added to the red solution, no colour change occurred.

Which of these substances, when added, is most likely to cause the red solution to change colour?

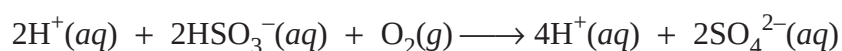
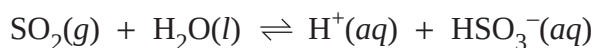
- (A) Cleaning solution containing ammonia
- (B) Concentrated hydrochloric acid
- (C) Orange juice
- (D) Vinegar

- 8 The burning of sulfur can be described by the following equation:



What volume of sulfur dioxide gas will be released at 25°C and 101.3 kPa when 8.00 g of sulfur is burnt?

- (A) 3.06 L
 - (B) 6.12 L
 - (C) 12.24 L
 - (D) 24.47 L
- 9 An understanding of Le Chatelier's principle is important in the chemical industry. Which prediction can be made using this principle?
- (A) The identity of products of a chemical reaction
 - (B) The effect of changes in temperature on the rates of reactions
 - (C) The effect of catalysts on the position of equilibrium reactions
 - (D) The effect of changes in the concentration of chemical substances in equilibrium
- 10 The following equations describe some reactions in the formation of acid rain:



What would occur if some solid sodium sulfate (Na_2SO_4) were added to a sample of acid rain?

- (A) The amount of $\text{SO}_2(\text{g})$ would increase and the acidity of the solution would decrease.
- (B) The amount of $\text{SO}_2(\text{g})$ would increase and the acidity of the solution would increase.
- (C) The amount of $\text{SO}_2(\text{g})$ would be unchanged and the acidity of the solution would be unchanged.
- (D) The amount of $\text{SO}_2(\text{g})$ would be unchanged and the acidity of the solution would decrease.

- 11 Why is chlorine used to treat local water supplies?
- (A) To make water suitable for swimming
(B) To kill micro-organisms living in the water
(C) To promote sedimentation of finely suspended solids
(D) To precipitate heavy metal ions such as lead and mercury
- 12 The atomic absorption spectrophotometer was developed by Sir Alan Walsh and his team at CSIRO in the 1950s. Its development was one of the most significant in Australian chemical technology. What did it provide?
- (A) A rapid method to monitor chemical pollutants in water supplies
(B) The first method for determining the concentrations of metal ions in water supplies
(C) A method for determining the concentrations of hydrocarbons at very low concentrations
(D) A method for determining the concentrations of metal ions at very low concentrations
- 13 Four students analysed a sample of fertiliser to determine its percentage of sulfate.
- Each student:
- weighed an amount of fertiliser;
 - dissolved this amount in 100 mL of water;
 - added aqueous barium nitrate;
 - filtered, dried and weighed the barium sulfate precipitate.

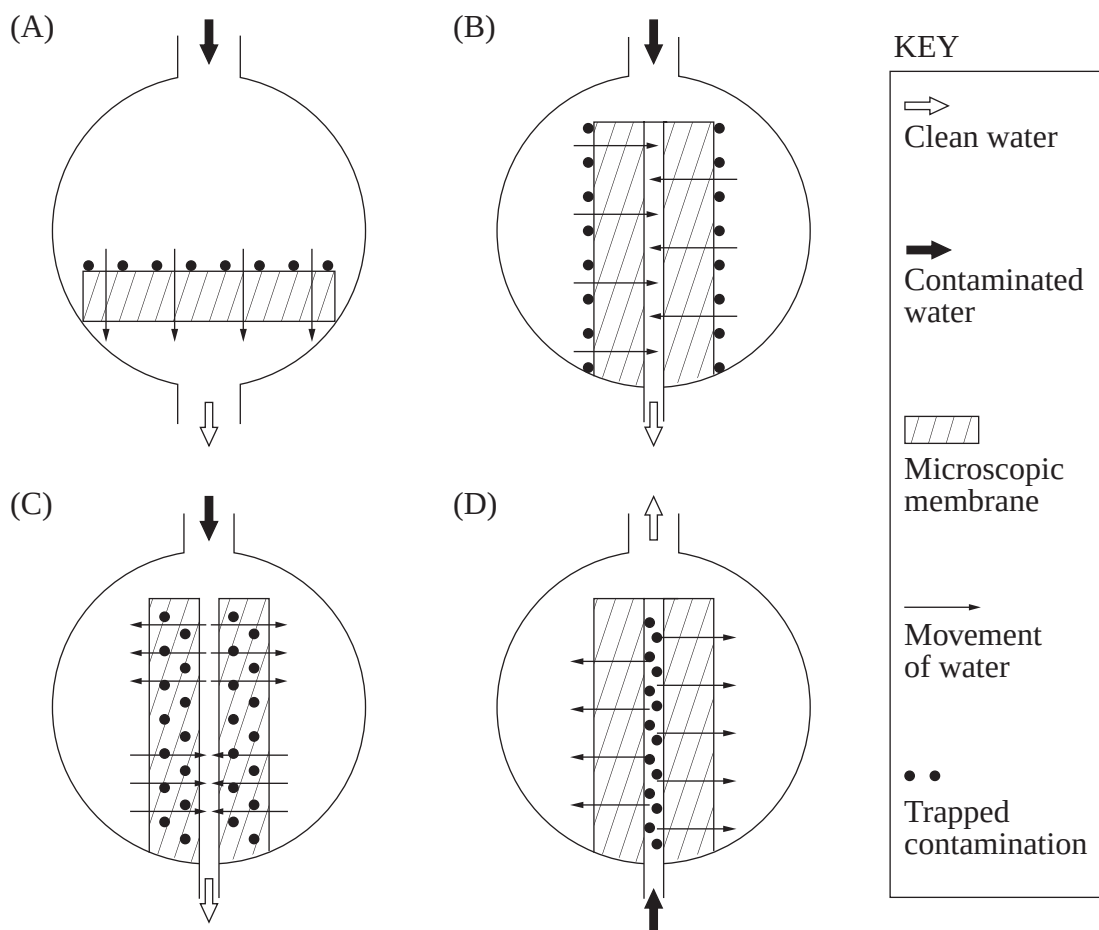
Their results and calculations are shown in the table.

<i>Student</i>	<i>Mass of fertiliser used (g)</i>	<i>Mass of BaSO₄ weighed (g)</i>	<i>Percentage of sulfate in fertiliser (%)</i>
<i>A</i>	11.6	19.5	69.2
<i>B</i>	10.4	16.9	66.9
<i>C</i>	10.268	22.612	90.6
<i>D</i>	11.1	18.2	67.5

The percentage of sulfate calculated by Student C was significantly higher than that of the other students. Which is the most likely reason for this?

- (A) Student C did not dry the sample for long enough.
(B) Student C added more Ba(NO₃)₂ solution than the other students.
(C) Student C used a balance capable of measuring weight to more decimal places.
(D) Student C waited longer than the other students for the Ba(NO₃)₂ to react completely with the sulfate.

- 14 Which diagram represents the most effective design for a microscopic membrane filter to purify contaminated water?



- 15 Four students were asked to test a solution for the presence of a cation by using various anions. The students obtained these results:

Student	Chloride	Sulfate	Carbonate
A	no precipitate	no precipitate	precipitate
B	precipitate	precipitate	no precipitate
C	precipitate	precipitate	precipitate
D	no precipitate	precipitate	no precipitate

Each student concluded that Pb^{2+} was present.

Which student had results consistent with this conclusion?

- (A) A
(B) B
(C) C
(D) D

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Centre Number

Section I (continued)

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Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Student Number

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (3 marks)

Radioisotopes are used in industry, medicine and chemical analysis. For ONE of these fields, relate the use of a named radioisotope to its properties.

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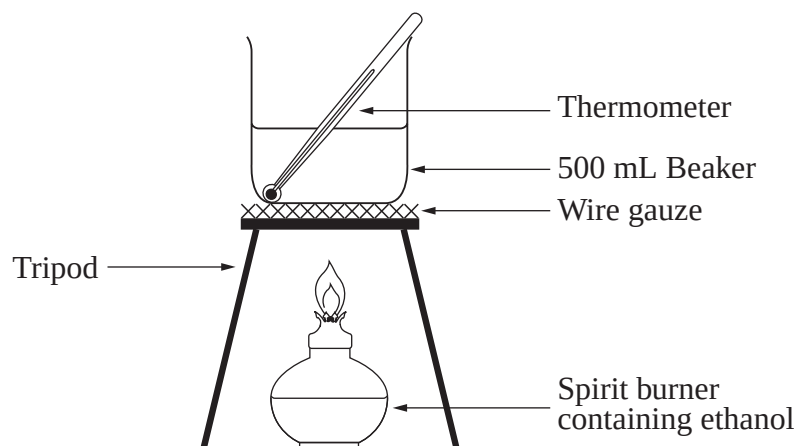
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Question 17 (6 marks)

Students were asked to perform a first-hand investigation to determine the molar heat of combustion of ethanol.

The following extract is from the practical report of one student.

Apparatus used:



Lab data:

Mass of water	=	250.0 g
Initial mass of burner	=	221.4 g
Final mass of burner	=	219.1 g
Initial temperature of water	=	19.0°C
Final temperature of water	=	59.0°C

- (a) After completing the calculations correctly, the student found that the answer did not agree with the value found in data books. Suggest ONE reason for this. 1

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- (b) Propose TWO adjustments that could be made to the apparatus or experimental method to improve the accuracy of the results. 2

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Question 17 continues on page 11

Question 17 (continued)

- (c) Calculate the molar heat of combustion of ethanol, using the student's data. **3**

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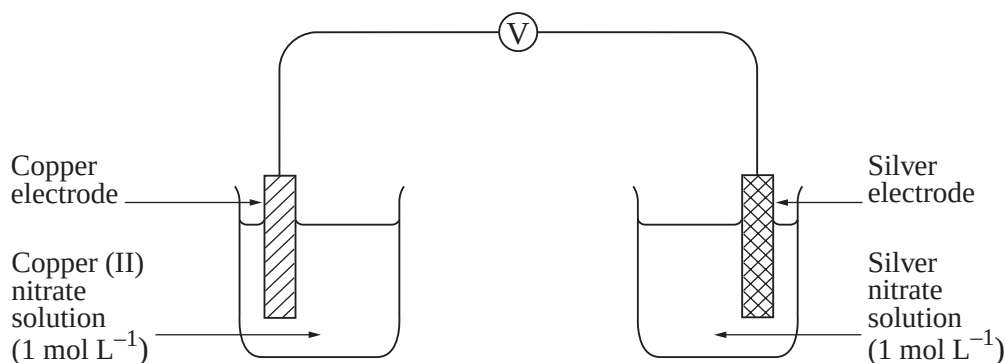
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End of Question 17

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Question 18 (6 marks)

A galvanic cell was made by connecting two half-cells. One half-cell was made by putting a copper electrode in a copper (II) nitrate solution. The other half-cell was made by putting a silver electrode in a silver nitrate solution. The electrodes were connected to a voltmeter as shown in the diagram.



(a) Complete the above diagram by drawing a salt bridge. 1

(b) Using the *standard potentials* table in the data sheet, calculate the theoretical voltage of this galvanic cell. 2

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(c) A student removes the voltmeter from the circuit and replaces it with an electrical generator. The generator causes the copper electrode to increase in mass. 3

Explain, using an equation, why the copper electrode will increase in mass.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 19** (7 marks)

Name ONE type of cell, other than the dry cell or lead–acid cell, you have studied. Evaluate it in comparison with either the dry cell or lead–acid cell, in terms of chemistry and the impact on society. Include relevant chemical equations in your answer.

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Question 20 (4 marks)

A 0.1 mol L^{-1} solution of hydrochloric acid has a pH of 1.0, whereas a 0.1 mol L^{-1} solution of citric acid has a pH of 1.6.

- (a) State ONE way in which pH can be measured. **1**

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- (b) Explain why the two solutions have different pH values. **3**

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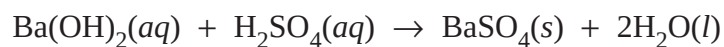
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Question 21 (4 marks)

Barium hydroxide and sulfuric acid react according to the following equation:

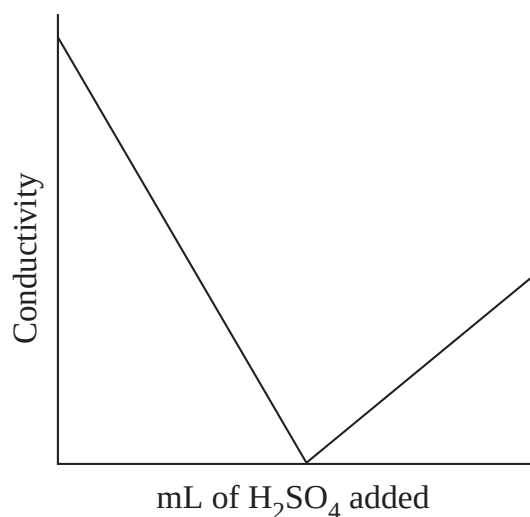


- (a) Name this type of chemical reaction.

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- (b) A 20 mL sample of barium hydroxide was titrated with 0.12 mol L^{-1} sulfuric acid. The conductivity of the solution was measured throughout the titration and the results graphed, as shown.

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Explain the changes in conductivity shown by the graph.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 22** (6 marks)

Justify the procedure you used to prepare an ester in a school laboratory. Include relevant chemical equations in your answer.

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Question 23 (4 marks)

A household cleaning agent contains a weak base of general formula NaX. 1.00 g of this compound was dissolved in 100.0 mL of water. A 20.0 mL sample of the solution was titrated with $0.1000 \text{ mol L}^{-1}$ hydrochloric acid and required 24.4 mL of the acid for neutralisation.

- (a) What is the Brønsted–Lowry definition of a base?

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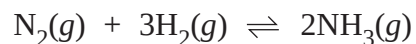
- (b) What is the molar mass of this base?

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Question 24 (6 marks)

In the early twentieth century, Fritz Haber developed a method for producing ammonia, as shown by the equation:



- (a) Ammonia is used as a cleaning agent. State ONE other use of ammonia. **1**

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- (b) Explain the effect of liquefying the ammonia on the yield of the reaction. **2**

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- (c) Explain why it is essential to monitor the temperature and pressure inside the reaction vessel. **3**

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Centre Number

Section I – Part B (continued)

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Student Number

Marks**Question 25** (6 marks)

Explain the need for monitoring the products of a chemical reaction such as combustion.

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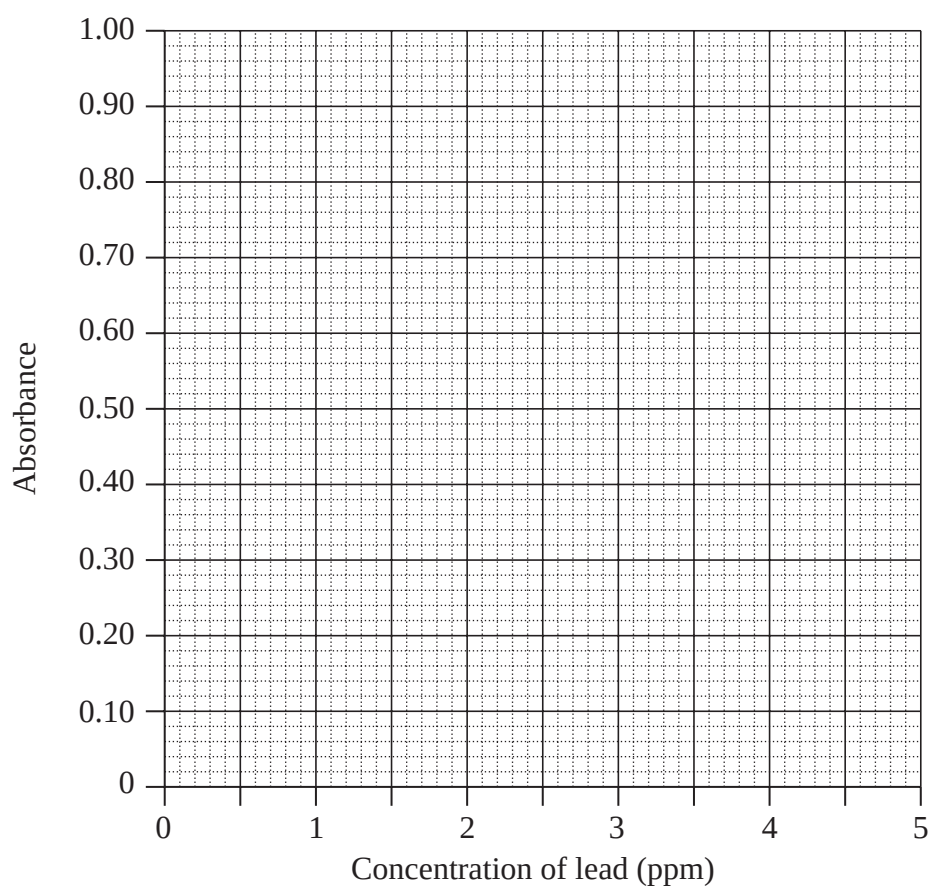
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Question 26 (4 marks)

A university student decided to measure the concentration of lead (Pb) in the soil around his home. He prepared five standard lead solutions of known concentration. The absorbance of these solutions was measured. These results are shown in the table.

<i>Concentration of lead standard (ppm)</i>	<i>Absorbance</i>
0	0.00
1	0.15
2	0.31
3	0.44
4	0.59
5	0.75

(a) Draw a line graph of these data.

1

Question 26 continues on page 23

Question 26 (continued)

- (b) The student prepared solutions from four different soil samples around his home. These solutions were also analysed using the same method. The results are shown in the table. 1

<i>Solutions made from soil samples</i>	
<i>Area sampled</i>	<i>Absorbance</i>
Front garden bed	0.19
Back garden bed	0.09
Mail box	0.22
Back fence	0.11

Determine the highest concentration of lead in the soil around the home.

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- (c) State an hypothesis to account for the variation in lead concentration around the student's home. 2

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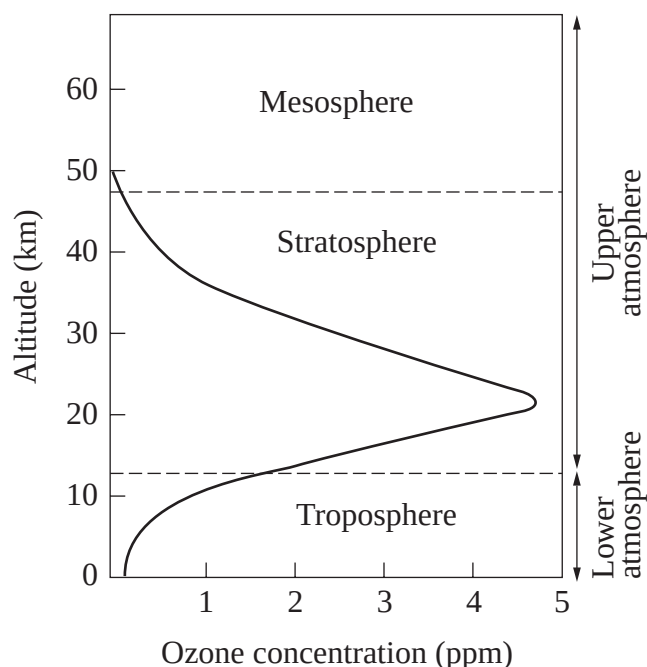
End of Question 26

Please turn over

Question 27 (4 marks)

Oxygen exists in the atmosphere as the allotropes oxygen and ozone. The graph shows a typical change in ozone concentration with changing altitude.

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Compare the environmental effects of the presence of ozone in the upper and lower atmosphere.

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 28–32

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

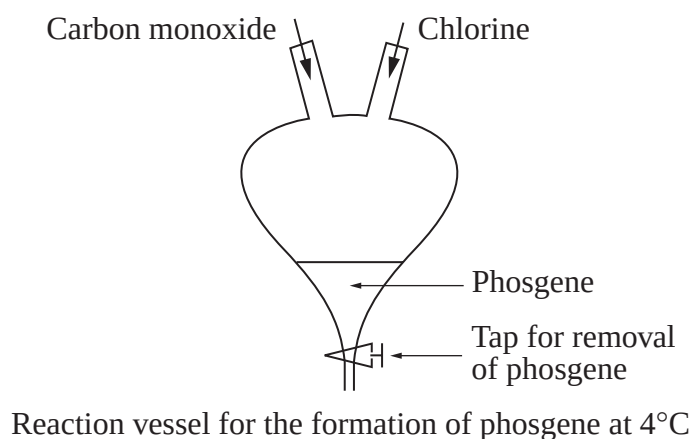
Show all relevant working in questions involving calculations.

	Pages
Question 28 Industrial Chemistry	26
Question 29 Shipwrecks and Salvage	27
Question 30 The Biochemistry of Movement	28
Question 31 The Chemistry of Art	29–30
Question 32 Forensic Chemistry	31

Question 28 — Industrial Chemistry (25 marks)

Electrolysis is an important industrial process.

- (a) (i) Define *electrolysis*. 1
- (ii) Compare the reaction products from the electrolysis of molten sodium chloride and concentrated aqueous sodium chloride. 2
- (b) Carbonyl chloride, COCl_2 , is a colourless, poisonous gas that is also known as phosgene. It is needed for the production of insecticides, polyurethane plastics and polycarbonate. It is produced from the exothermic equilibrium reaction of carbon monoxide gas and chlorine gas. When the reaction vessel is cooled below 8°C the phosgene is a liquid.

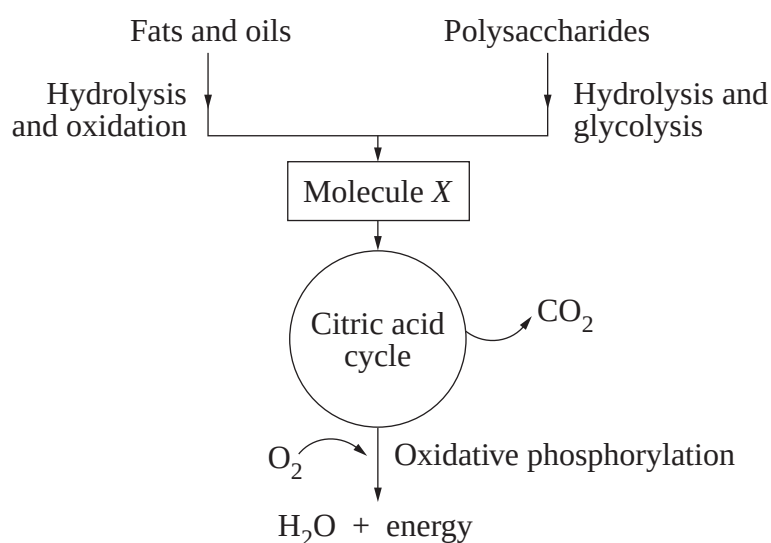


- (i) Write a balanced equation for the formation of phosgene. 2
- (ii) Explain how industry could maximise the production of phosgene. 2
- (c) Explain why sulfuric acid is an important industrial chemical. Include balanced chemical equations in your answer. 5
- (d) (i) Name the chemical process used to make soap. 1
- (ii) Outline the procedure for making soap in the school laboratory. 2
- (iii) Describe a safety risk associated with the procedure outlined in part (ii), and suggest a safe work practice to minimise the risk. 3
- (e) Evaluate how environmental issues are addressed in the Solvay process. 7

			Marks
Question 29 — Shipwrecks and Salvage (25 marks)			
(a)	(i)	Identify the main metal used to construct ships.	1
	(ii)	Although aluminium is a very reactive metal, with a very low reduction potential, it is used in many structures exposed to oxidising conditions. Explain why aluminium can be used in this way.	2
(b)	(i)	Give an example of a metal commonly used as a sacrificial anode.	1
	(ii)	Explain why sacrificial anodes are added to metal-hulled ships.	3
(c)		Describe the effect of adding other elements to iron on the properties and uses of steels.	5
(d)	(i)	Define <i>corrosion</i> .	1
	(ii)	Outline a procedure that could be used to compare the corrosion rates of different metals or alloys in the school laboratory.	2
	(iii)	Describe ways in which accuracy and reliability could be improved in the procedure described in part (ii).	3
(e)		For ONE specific metal, evaluate the steps that can be used to clean, stabilise and preserve artefacts recovered from shipwrecks.	7

Question 30 — The Biochemistry of Movement (25 marks)

- (a) (i) Name the molecule that stores energy for nearly all metabolic processes. **1**
- (ii) Explain how the biologically important part of the molecule in part (i) provides energy for cellular metabolism. **2**
- (b) Energy for our bodies to function is provided by the oxidation of fuels. Fats and glucose are used as fuels. A summary of the oxidation of fats and glucose is shown.

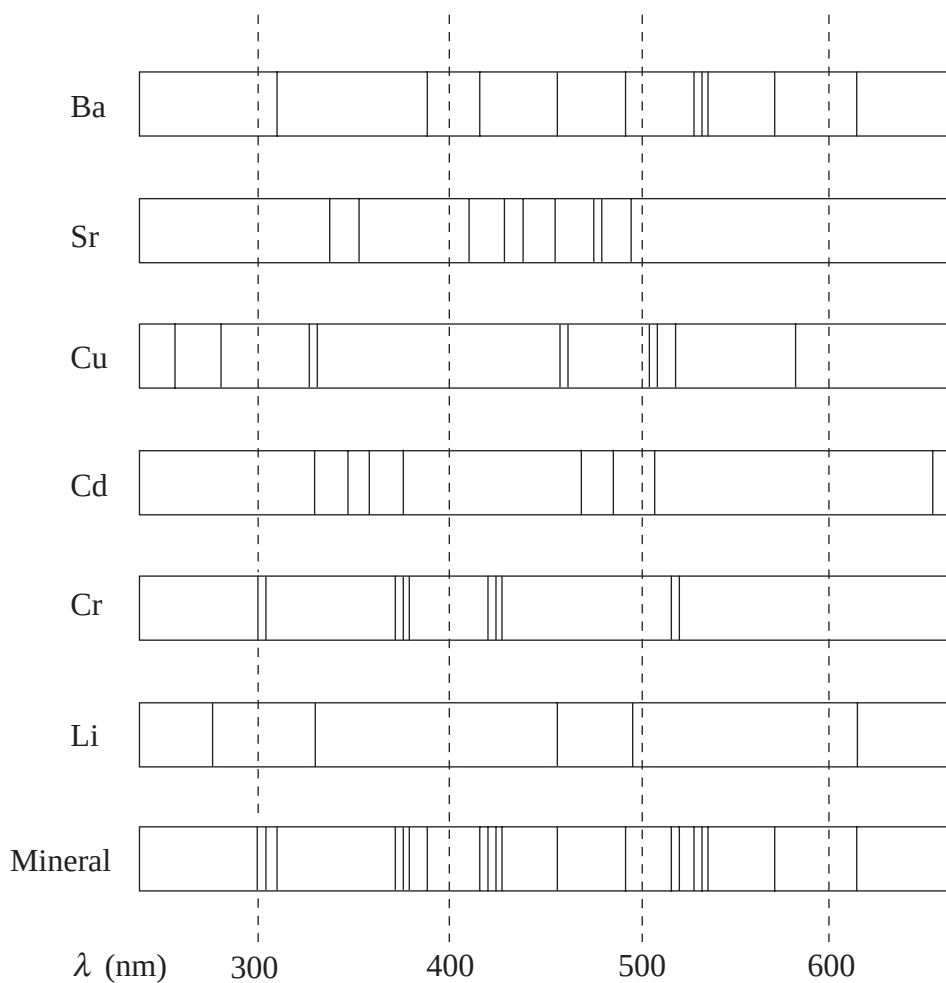


- (i) Identify molecule *X* and state its function. **2**
- (ii) Analyse the role of oxidative phosphorylation in energy production. **2**
- (c) Discuss the use of models in developing an understanding of enzyme function. **5**
- (d) (i) Define *viscosity*. **1**
- (ii) Outline a procedure that could be used to compare the viscosity of pure glycerol and a glycerol solution. **2**
- (iii) Describe ways in which accuracy and reliability could be improved in the procedure described in part (ii). **3**
- (e) The energy requirements for different types of skeletal muscle are met by the interaction of separate energy output systems. **7**

Analyse the role and interaction of the energy output systems used by skeletal muscle.

Question 31 — The Chemistry of Art (25 marks)

- (a) (i) Identify the mineral source of a pigment. 1
- (ii) Describe the use of a named separation process to obtain a pigment. 2
- (b) A sample of a mineral used as a pigment was subjected to spectroscopic analysis. The spectrum is shown.



- (i) Identify the elements in the mineral sample. 2
- (ii) Explain how a line spectrum is produced. 2

Question 31 continues on page 30

	Marks
Question 31 (continued)	
(c) Describe the Bohr model of the atom, and identify ONE merit and ONE limitation of the model.	5
(d) (i) Name a transition element.	1
(ii) Outline a first-hand investigation to demonstrate the colour changes of a named transition element as it changes in oxidation state.	2
(iii) Describe a safety risk associated with the procedure outlined in part (ii), and suggest a safe work practice to minimise the risk.	3
(e) With reference to TWO pigments, explain how the metallic components produce colour.	7

End of Question 31

Question 32 — Forensic Chemistry (25 marks)

- (a) (i) Define *organic compounds*. 1
- (ii) For ONE class of organic compound, describe a chemical test that identifies this class. 2

- (b) The table shows fatty acid composition of some common oils and fats.

<i>Fatty acids present (% by weight)</i>					
Fat or oil hydrolysed	Lauric	Palmitic	Stearic	Oleic	Linoleic
Butter	2–3	23–26	10–13	30–40	4–5
Lard	< 1	28–30	12–18	41–48	6–7
Tallow	< 1	24–32	14–32	35–38	2–4
Coconut	45–51	4–10	1–5	2–10	0–2

An oily sample was hydrolysed and the fatty acids analysed.

<i>Fatty acids present (% by weight)</i>					
	Lauric	Palmitic	Stearic	Oleic	Linoleic
Oil sample	< 1	29	28	36	4

- (i) Which fat or oil has been identified? 1
- (ii) Explain the solubility in water of fatty acids, in terms of their structure. 3
- (c) Assess the usefulness of mass spectrometry in providing forensic evidence. 5
- (d) (i) Name ONE technique used by forensic chemists to separate a mixture of organic compounds. 1
- (ii) In your study of Forensic Chemistry, you performed a first-hand investigation to separate a mixture of organic materials. Outline the procedure. 2
- (iii) Describe ways in which accuracy and reliability could be improved in the procedure described in part (ii). 3
- (e) Discuss the uses of DNA analysis in forensic chemistry. 7

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DATA SHEET

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 101.3 kPa (1.00 atm) and	
at 273 K (0°C)	22.41 L
at 298 K (25°C)	24.47 L
Ionisation constant for water at 298 K (25°C), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

KEY									
Atomic Number		Symbol of element		Atomic Weight		Name of element			
79		Au		197.0		Gold			
Atomic Number		Symbol of element		Atomic Weight		Name of element			
1		H		1.008		Hydrogen			
3		Li		6.941		Lithium			
4		Be		9.012		Beryllium			
11		Na		22.99		Sodium			
12		Mg		24.31		Magnesium			
20		Ca		40.08		Calcium			
21		Sc		44.96		Scandium			
39		Y		88.91		Yttrium			
38		Sr		87.62		Strontium			
56		Ba		137.3		Barium			
57-71		Lanthanides							
88		Ra		[226.0]		Radium			
87		Fr		[223.0]		Francium			
55		Cs		132.9		Caesium			
72		Hf		178.5		Hafnium			
73		Ta		180.9		Tantalum			
104		Rf		[261.1]		Rutherfordium			
105		Db		[262.1]		Dubnium			
106		Sg		[263.1]		Seaborgium			
107		Bh		[264.1]		Bohrium			
108		Hs		[265.1]		Hassium			
109		Mt		[268]		Meitnerium			
110		Uun		—		Ununnilium			
111		Uuu		—		Unununium			
112		Uub		—		Ununbium			
113									
81		Tl		204.4		Thallium			
82		Pb		207.2		Lead			
83		Bi		209.0		Bismuth			
84		Po		[210.0]		Polonium			
85		At		[210.0]		Astatine			
53		I		126.9		Iodine			
52		Te		127.6		Tellurium			
34		Se		78.96		Selenium			
35		Br		79.90		Bromine			
36		Kr		83.80		Krypton			
33		As		74.92		Arsenic			
51		Sb		121.8		Antimony			
54		Xe		131.3		Xenon			
116		Uuh		—		Ununhexium			
117									
118		Uuo		—		Ununoctium			
2		He		4.003		Helium			

Lanthanides

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
138.9	140.1	140.9	144.2	[146.9]	150.4	152.0	157.3	158.9	162.5	164.9	167.3	168.9	173.0	175.0
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium

Actinides

89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
[227.0]	232.0	231.0	238.0	[237.0]	[239.1]	[241.1]	[244.1]	[249.1]	[252.1]	[252.1]	[257.1]	[258.1]	[259.1]	[262.1]
Actinium	Thorium	Protactinium	Uranium	Nepunium	Plutonium	Americium	Curtum	Berkelium	Californium	Einsteinium	Fermium	Mendelevium	Nobelium	Lawrencium

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.